
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	RWASHANDE, Seith
Date of birth:	2/4/1990
Subject:	020-AGK
Result:	34,99 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2763		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	3651		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	3619		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	2787		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
5	2910		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
6	2414	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
7	2418	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
8	2918	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
9	2449	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
10	615	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
11	80	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
12	1043	21.8.2.2	Design, operation, system components, etc	1,00	1,00
13	1074	21.8.2.2	Design, operation, system components, etc	1,00	1,00
14	443	21.9.1	General, definitions, basic applications: etc	1,00	1,00
15	926	21.9.1.6	Electromagnetism	0,00	1,00
16	913	21.9.1.7	Circuit breakers	0,00	1,00
17	1286	21.9.3.2	AC Generation	1,00	1,00
18	1449	21.10.1.2	Engine: design, operation etc	0,00	1,00
19	198	21.10.1.2	Engine: design, operation etc	0,00	1,00
20	672	21.11.2.1	Aeroplane: Air intake	0,00	1,00
21	4028		22. INSTRUMENTATION	1,00	1,00
22	3107		22. INSTRUMENTATION	0,00	1,00
23	3100		22. INSTRUMENTATION	0,00	1,00
24	2797		22. INSTRUMENTATION	0,00	1,00
25	3424		22. INSTRUMENTATION	0,00	1,00
26	4025		22. INSTRUMENTATION	0,00	1,00
27	4022		22. INSTRUMENTATION	1,00	1,00
28	2806		22. INSTRUMENTATION	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	4464	22.	INSTRUMENTATION	0,00	1,00
30	3112	22.	INSTRUMENTATION	0,00	1,00
31	2824	22.	INSTRUMENTATION	1,00	1,00
32	3102	22.	INSTRUMENTATION	1,00	1,00
33	4026	22.	INSTRUMENTATION	0,00	1,00
34	3090	22.	INSTRUMENTATION	0,00	1,00
35	2822	22.	INSTRUMENTATION	0,00	1,00
36	34	22.	INSTRUMENTATION	1,00	1,00
37	694	22.2.4	Altimeter	1,00	1,00
38	1144	22.2.6	Airspeed indicator (ASI)	1,00	1,00
39	1140	22.2.6	Airspeed indicator (ASI)	1,00	1,00
40	1711	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
Total:35%				14,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: RWASHANDE, Seith

Location: Uganda Civil Aviation Authority

Your Result: 14,00 from 40,00 Points (35,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	6,00	20,00	30,00
21.2. AIRFRAME	4	1,00	4,00	25,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	2	1,00	2,00	50,00
21.8. FUEL SYSTEM	2	2,00	2,00	100,00
21.9. ELECTRICS	4	2,00	4,00	50,00
21.10. PISTON ENGINES	2	0,00	2,00	0,00
21.11. TURBINE ENGINES	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	8,00	20,00	40,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	3	3,00	3,00	100,00
22.4. GYROSCOPIC INSTRUMENTS	1	0,00	1,00	0,00
Total	40	14,00	40,00	35,00

1

The low temperature that causes carburetor ice in an engine equipped with a float-type carburetor is normally the result of the

- ☒ compression of air at the carburetor venturi.
- ☐ vaporization of fuel and expansion of air in the carburetor.
- ☐ freezing temperature of the air entering the carburetor.

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2

The purpose of a kill switch is to

- ☐ ground the lead wire to the ignition coil shutting down the powerplant.
- ☐ ground the battery eliminating current for the ignition system.
- ☒ shut off the fuel to the carburetor.

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3

The uncontrolled firing of the fuel/air charge in advance of normal spark ignition is known as

☐ detonation.

☐ pre-ignition.

☒ instantaneous combustion.

4

If the ground wire between the magneto and the ignition switch becomes disconnected, the most noticeable result will be that the engine

- ☐ cannot be shut down by turning the switch to the OFF position.
- ☐ cannot be started with the switch in the ON position.
- ☒ will run very rough.

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5

What type of oxygen should be used to replenish an aircraft oxygen system?

☒ **Medical.**

☐ **Industrial.**

☐ **Aviation.**

6

What effect does an increase in airspeed have on a coordinated turn while maintaining a constant angle of bank and altitude?

- ☒ **The rate of turn will increase resulting in an increased load factor.**
- ☐ **The rate of turn will decrease resulting in no changes in load factor.**
- ☐ **The rate of turn will decrease resulting in a decreased load factor.**

7

If no corrective action is taken by the pilot as angle of bank is increased, how is the vertical component of lift and sink rate affected?

- ☐ Lift decreases and the sink rate decreases.
- ☐ Lift increases and the sink rate increases.
- ☒ Lift decreases and the sink rate increases.

8

How should a balloon fuel system be checked for leaks prior to flight?

- ☒ Check for unusual discoloration on burner coils.
- ☐ Look, listen, and smell.
- ☐ Cover all fittings and hoses with soapy water and look for small bubbles.

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9

By changing the angle of attack of a wing, the pilot can control the airplane's

☒ lift, airspeed, and drag.

☐ lift and airspeed, but not drag.

☒ lift, gross weight, and drag.

10

What can be the consequence during a descent with a fully open throttle if the waste gate is seized ?

- ☐ The manifold air pressure (MAP) value may exceed the maximum allowed value.
- ☐ The turbine shaft will break.
- ☒ The power of the motor will decrease.
- ☐ The turbine blades will separate.

11

At constant fuel flow, if engine compressor air is bled off for engine anti-icing or a similar system, the turbine temperature:

- ☒ will rise.
- ☐ may rise or fall depending on which stage of the compressor is used
- ☐ will fall.
- ☐ will be unchanged.

12

The pressurization of tanks is maintained by the fuel:

- ☐ tank drains.
- ☐ dump system.
- ☒ vent system.
- ☐ top off unit.

13

The main purpose of the mixture control is to:

- ☐ increase the oxygen supplied to the engine
- ☒ adjust the fuel flow to obtain the proper fuel/air ratio
- ☐ decrease the air supplied to the engine
- ☐ decrease oxygen supplied to the engine

14

On modern transport aircraft, cockpit windows are protected against icing by :

- ☐ Anti-icing fluid.
- ☒ Electric heating.
- ☐ Vinyl coating.
- ☐ Rain repellent system.

15

If a current is passed through a conductor,

- ☐ the intensity of the magnetic field will decrease.
- ☒ a force will be exerted on the conductor.
- ☐ there will be no effect unless the conductor is moved.
- ☒ the current will increase.

16

The function of the Generator Breaker is to close when the voltage of the :

- ☐ alternator is greater than the battery voltage and to open
- ☒ generator is greater than battery voltage
- ☐ battery is greater than the generator voltage and to open
- ☒ battery is greater than the alternator voltage and to open

17

A unit that converts electrical DC into AC is :

- ☐ a thermistor.
- ☐ an AC generator.
- ☐ a transformer rectifier unit.
- ☒ an inverter.

18

When changing power on engines equipped with constant-speed propeller, engine overload is avoided by accidental depressurization.

- ☐ increasing the RPM before increasing the manifold pressure.
- ☒ adjusting Fuel Flow before the manifold pressure.
- ☐ increasing the manifold pressure before increasing the RPM.
- ☐ reducing the RPM before reducing the manifold pressure.

19

The part of a piston engine that transforms reciprocating movement into rotary motion is termed as the :

☐ reduction gear

☐ piston

☒ camshaft

☐ crankshaft

20

The useful work area in an ideal Otto engine indicator diagram is enclosed by the following gas state change lines

- ☐ 2 adiabatic and 2 isobaric lines.
- ☐ 2 adiabatic, 1 isochoric and 1 isobaric lines.
- ☒ 2 adiabatic and 1 isothermic lines.
- ☐ 2 adiabatic and 2 isochoric lines.

21

When making an airborne VOR check, what is the maximum allowable tolerance between the two indicators of a dual VOR system (units independent of each other except the antenna)?

- ☒ **4 degrees between the two indicated bearings of a VOR.**
- ☐ **6 degrees between the two indicated radials of a VOR.**
- ☐ **Plus or minus 4 degrees when set to identical radials of a VOR.**

22

What are the indications of the pulsating VASI?

- ☒ High - pulsing white, on glidepath - steady white, slightly below glide slope steady red, low - pulsing red.
- ☐ High - pulsing white, on course and on glidepath - steady white, off course but on glidepath - pulsing white and red; low - pulsing red.
- ☒ High - pulsing white, on glidepath - green, low - pulsing red.

23

Identify the runway distance remaining markers.

- ☒ Signs with increments of 1,000 feet distance remaining.
- ☐ Yellow marker laterally placed across the runway with signs on the side denoting distance to end.
- ☒ Red markers laterally placed across the runway at 3,000 feet from the end.

24

Deviation error of the magnetic compass is caused by

- ☐ certain metals and electrical systems within the aircraft.
- ☒ the difference in location of true north and magnetic north.
- ☐ northerly turning error.

25

Which procedure is recommended to prevent or overcome spatial disorientation?

- ☒ Rely entirely on the indications of the flight instruments.
- ☐ Avoid steep turns and rough control movements.
- ☒ Reduce head and eye movements to the greatest extent possible.

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26

In which publication can the VOR receiver ground checkpoint(s) for a particular airport be found?

☐ Aeronautical Information Manual.

☐ Airport/Facility Directory.

☒ En Route Low Altitude Chart.

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27

(Refer to Figure 82.) Which is an acceptable range of accuracy when performing an operational check of dual VOR's using one system against the other?

☐ 2

☒ 4

☐ 1

See Attachments:

1. instrument_82.jpg

figure 82

(Attachment
No.1)

28

Which statement is true about magnetic deviation of a compass?

- ☐ Deviation is the same for all aircraft in the same locality.
- ☒ Deviation is different in a given aircraft in different localities.
- ☐ Deviation varies for different headings of the same aircraft.

29

(Refer to Figures 139 and 141.) Which displacement from the localizer centerline and glide slope at the 1,300-foot point from the runway is indicated?

- ☒ **28 feet above the glide slope and approximately 250 feet to the left of the runway centerline.**
- ☐ **21 feet above the glide slope and approximately 320 feet to the left of the runway centerline.**
- ☐ **21 feet below the glide slope and approximately 320 feet to the right of the runway centerline.**

See Attachments:

- | | | |
|-----------------------|--------------------|-------------------|
| 1. instrument_139.jpg | instrument_139.jpg | (Attachment No.2) |
| 2. instrument_140.jpg | instrument_140.jpg | (Attachment No.3) |

30

What is the advantage of a three-bar VASI?

- ☒ The three-bar VASI is much more visible and can be used at a greater height.
- ☐ A normal glide angle is afforded both high and low cockpit aircraft.
- ☐ Pilots have a choice of glide angles.

31

Prior to starting the engine, the manifold pressure gauge usually indicates approximately 29" Hg. This is because the

- ☐ pointer on the gauge is stuck at the full-power indication.
- ☒ pressure within the manifold is the same as atmospheric pressure.
- ☐ throttle is closed, trapping high air pressure in the manifold.

32

Which may lead to a power push-over in a gyroplane?

- ☒ Rotor force is removed.
- ☐ Low speed.
- ☐ Decreasing power too quickly.

33

Which is the maximum tolerance for the VOR indication when the CDI is centered and the aircraft is directly over the airborne checkpoint?

- ☒ Plus or minus 4 degrees of the designated radial.
- ☐ Plus 6degrees or minus 4 degrees of the designated radial.
- ☐ Plus or minus 6 degrees of the designated radial.

34

Identify taxi leadoff lights associated with the centerline lighting system.

- ☒ **Alternate green and yellow lights curving from the centerline of the runway to the edge of the taxiway.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to a point on the exit.**
- ☐ **Alternate green and yellow lights curving from the centerline of the runway to the centerline of the taxiway.**

35

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☐ altimeter may indicate an altitude higher than the actual altitude being flown.
- ☒ altimeter may indicate an altitude lower than the actual altitude being flown.
- ☐ airspeed indicator may indicate less than normal.

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36

If the Pitot head is blocked, what airspeed indication can he expected?

- ☐ A decrease of IAS during a climb.
- ☐ Constant IAS during a descent.
- ☒ No change of IAS in level flight, even with large power changes.

37

The high Altitude Radio Altimeter uses the following wavelengths:

☐ **metric.**

☐ **hectometric.**

☒ **decimetric.**

☐ **myriametric.**

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38

During a climb, the static line becomes blocked, the ASI will:

- ☒ under-read
- ☐ read zero
- ☐ read normally
- ☐ over-read

39

An ASI consists of a gas tight casing which has:

- ☐ static pressure fed through a restrictive choke
- ☐ static pressure fed outside a sealed capsule and pitot pressure fed into a second open capsule
- ☒ pitot pressure fed into an open capsule and static pressure fed outside the capsule
- ☐ static pressure fed outside a sealed capsule

40

During a stabilised climbing turn at a constant rate, the instruments which indicate the correct pitch and bank are the:

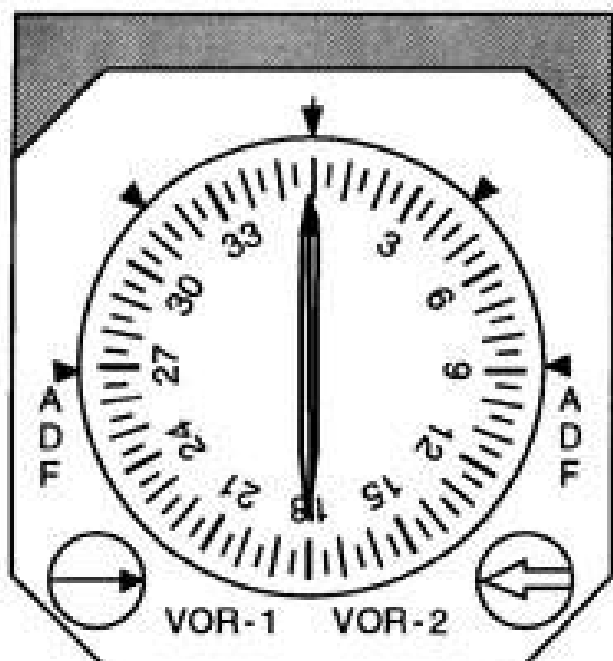
- ☐ vertical-speed indicator and turn-and-slip indicator;
- ☐ attitude indicator and turn-and-slip indicator.
- ☒ altimeter and turn-and-slip indicator;

Attachment No. 1

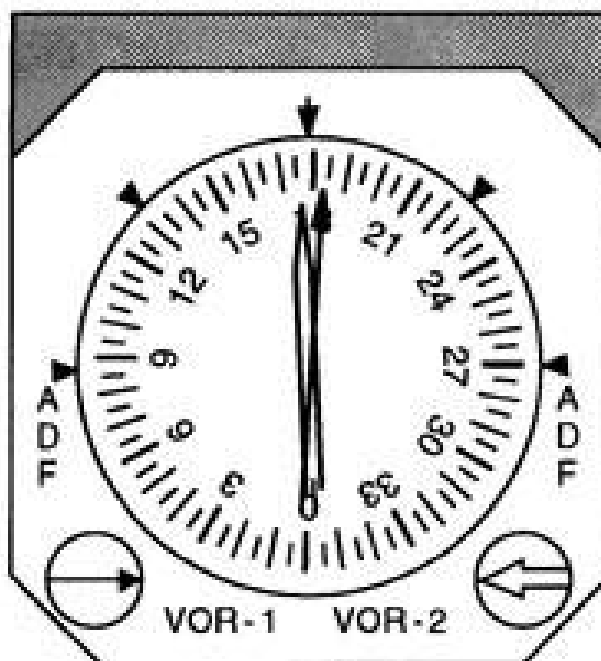
Questions 27

figure 82

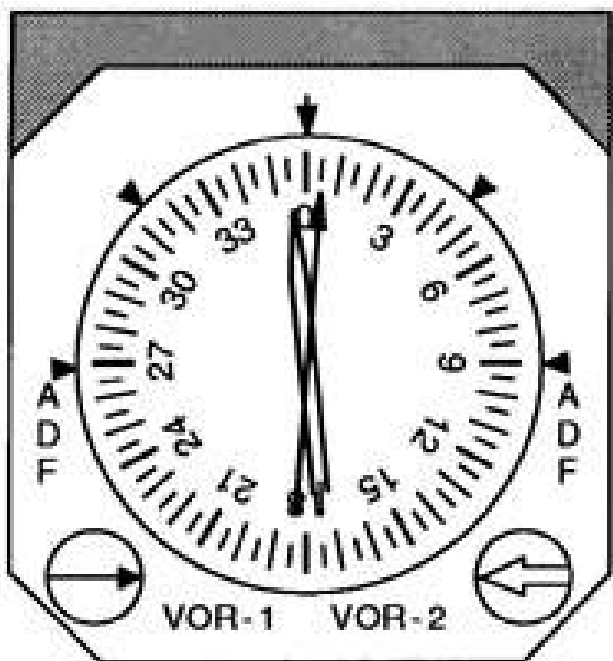
instrument_82.jpg



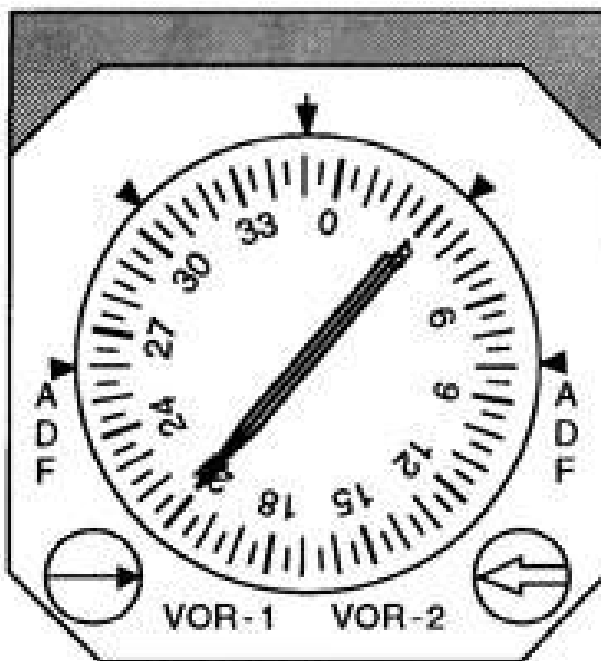
1



2



3



4

Figure 82. Dual VOR System, Accuracy Check © ASA

Attachment No. 2

Questions 29

instrument_139.jpg

instrument_139.jpg

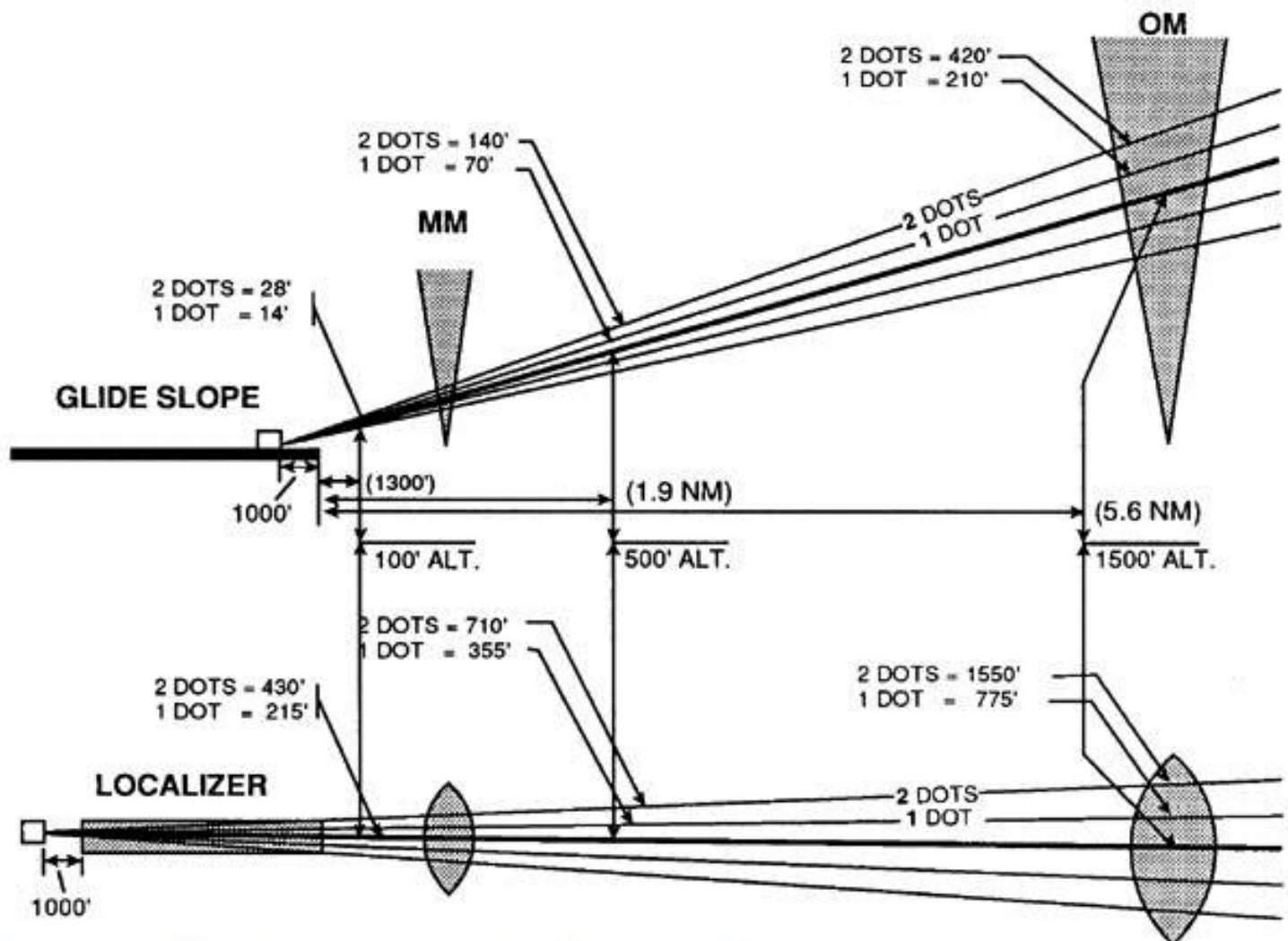


Figure 139. Glide Slope and Localizer Illustration © ASA

Attachment No. 3

Questions 29

instrument_140.jpg

instrument_140.jpg

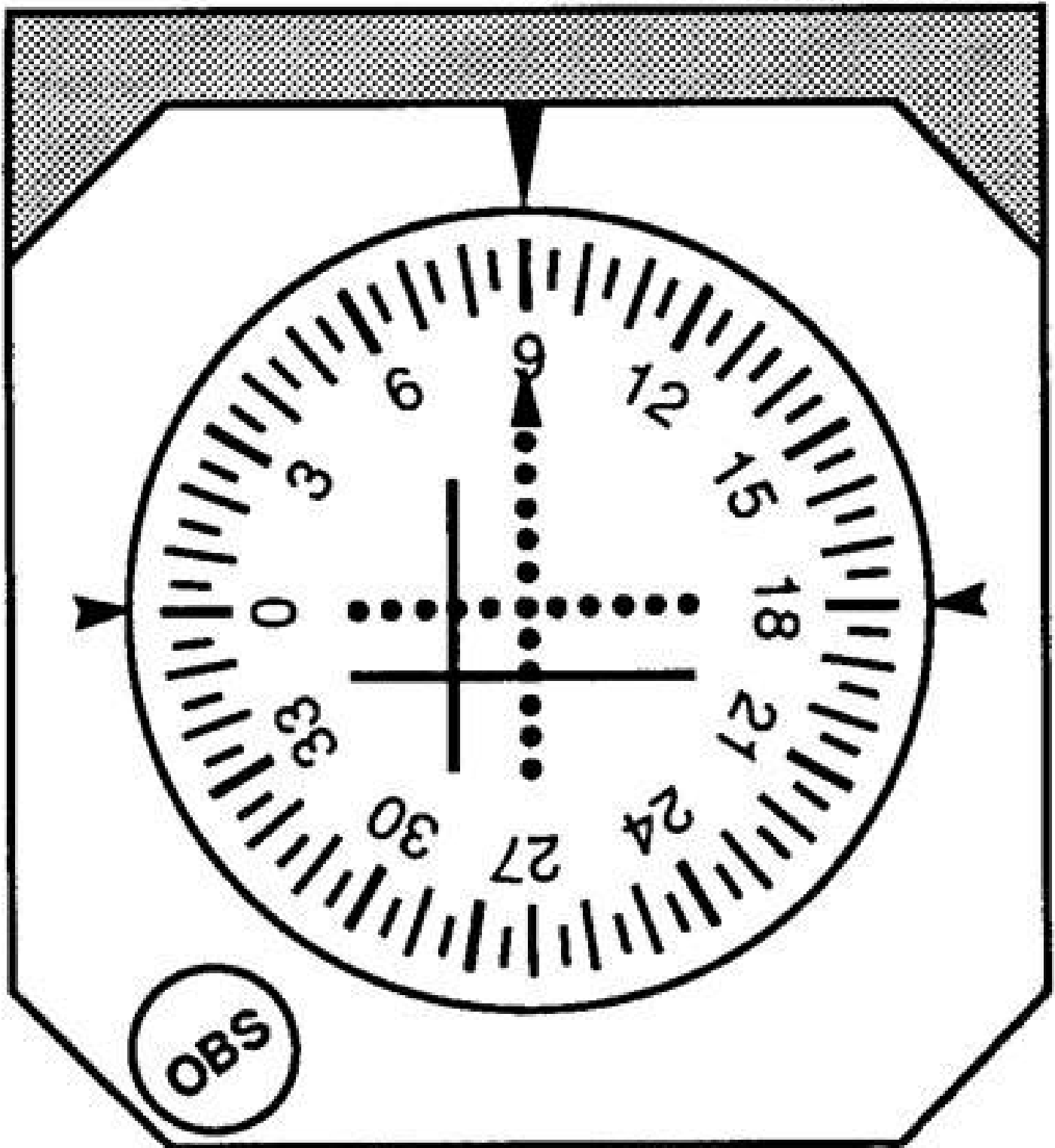


Figure 140. OBS, ILS, and GS Displacement © ASA